

Charotar University of Science & Technology (CHARUSAT)
Fourth Semester of B.Tech.(EC) Theory Examination (April-May 2018)
MA 203 Engineering Mathematics-IV

Date: 30/04/2018, Monday **Time:** 10:00 a.m. to 01:00 p.m **Maximum Marks:** 70

INSTRUCTIONS:

1. The question paper comprises two sections.
 2. Section I and II must be attempted in separate answer sheets.
 3. Make suitable assumptions and draw neat figures wherever required.
 4. Use of scientific calculator is allowed.
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SECTION-I

Q.1. Choose the correct answer for the following questions.

[07]

1. $\int_{|z|=5} \frac{z}{(z-7)(z+9)} dz = \underline{\hspace{2cm}}.$

- a. 0 b. 1 c. 7 d. 9.

2. A function $f(z) = |z|^2$ is $\underline{\hspace{2cm}}.$

- a. an entire function c. analytic only at origin
b. nowhere analytic d. analytic only at $z = 1$.

3. The mapping $w = z + c$, where c is complex constants is called $\underline{\hspace{2cm}}.$

- a. inversion c. rotation
b. translation d. magnification.

4. If $F(s)$ is the Fourier transform of function $f(x)$, then $\underline{\hspace{2cm}}.$

- a. $f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s) e^{-isx} ds$ c. $f(x) = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} F(s) e^{-isx} ds$
b. $f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s) \cos sx ds$ d. $f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s) \sin sx ds$

5. If the Laurent series of function $f(z)$ is $\frac{1}{z^2} + \frac{1}{z^3} + \frac{1}{2z^4} + \dots$, then $\underline{\hspace{2cm}}.$

- a. $f(z)$ has removable singularity at $z = 0$
b. $f(z)$ has essential singularity at $z = 0$
c. $f(z)$ has double pole at $z = 0$
d. $f(z)$ has simple pole at $z = 0$

6. Which of the following is true for a complex function?
- every differentiable function is an entire function
 - every continuous function is differentiable
 - every entire function is differentiable
 - every continuous function is an entire.

7. Which of the following function is harmonic?

- a. $x^2 - y^2$ b. $x^2 + y^2$ c. $x + y^2$ d. $x^2 + y$.

Q.2 Attempt any four questions.

[16]

- Expand $f(z) = \frac{1}{(z-1)(z+5)}$ in Laurent series valid for (a) $|z| < 1$, (b) $1 < |z| < 5$, (c) $|z| > 5$.
- Using residue theorem, evaluate the integral $\int_C \frac{4-3z}{z(z-1)(z-2)} dz$, where C is the circle $|z| = \frac{3}{2}$.
- Show that the function $u(x, y) = x^4 - 6x^2y^2 + y^4$ is harmonic. Also find the analytic function $f(z) = u(x, y) + iv(x, y)$.
- Discuss the analyticity of function $f(z) = z^2 + z - 1$.
- Find the image of $|z - 3i| = 3$ under the mapping $w = \frac{1}{z}$.
- Define Fourier transform of function $f(x)$. Find the Fourier transform of the function
$$f(x) = \begin{cases} 1 - x^2, & \text{for } -1 < x < 1, \\ 0, & \text{for } |x| > 1. \end{cases}$$

Q.3(a) Attempt any two questions.

[06]

- Find the Fourier cosine transform of
$$f(x) = \begin{cases} x, & \text{for } 0 < x < 1, \\ 2 - x, & \text{for } 1 < x < 2, \\ 0, & \text{for } x > 2. \end{cases}$$
- Define Fourier sine transform. Find the Fourier sine transform of e^{-5x} .
- Find a bilinear transformation which maps the points $i, -i, 1$ of the z -plane into $0, 1, \infty$ of the w -plane respectively.

(b) Attempt any two of the following questions. [06]

- (i) Show that a function e^{2z} is an entire function.
- (ii) State Cauchy's integral formula. Hence evaluate the integral $\oint_{|z|=2} \frac{e^{-z}}{z+1} dz$.
- (iii) Expand $\cos z$ in a Taylor Series about $z = \frac{\pi}{4}$.

SECTION-II

Q.4 Choose the correct answer for the following questions. [07]

1. In application of Simpsons $\frac{1}{3}$ rule, the interval of integration for closer approximation should be _____.
 - a. odd number of sub-intervals with small step-length
 - b. Even number of sub-intervals with small step-length
 - c. Even number of sub-intervals with large step-length
 - d. odd number of sub-intervals with large step-length
2. The relation between E and ∇ is _____.
 - a. $\nabla = 1 - E^{-1}$
 - b. $\nabla = 1 + E^{-1}$
 - c. $\nabla = 1 - E$
 - d. none of these.
3. The iterative formula of Euler's method for solution $\frac{dy}{dx} = f(x, y)$ with $y(x_0) = y_0$ is _____.
 - a. $y_{k+1} = y_k - h * f(x_k, y_k), k = 0, 1, 2, 3...$
 - b. $y_{k+1} = y_0 - h * f(x_k, y_k), k = 0, 1, 2, 3...$
 - c. $y_{k+1} = y_k + h * f(x_k, y_k), k = 0, 1, 2, 3...$
 - d. $y_{k+1} = y_k - h * f(x_k, y_0), k = 0, 1, 2, 3...$
4. Assume we have a table of values $(x_i, y_i), i = 0, 1, 2, 3$ of a function $y = f(x)$. One can obtain Lagrange interpolating polynomial of degree _____.
 - a. 1
 - b. 2
 - c. at most 3
 - d. 3
5. Assume that Gauss Jacobi method is used to solve the system of equations

$$10x + y - 2z = 17, \quad 3x + 10y - z = -18, \quad 2x - 3y + 10z = 25$$

with initial guess $(0, 0, 0)$. Then the approximate solution to the system after first iteration is _____.

a. $x = 17, y = 0, z = 0$

b. $x = \frac{17}{10}, y = \frac{-18}{10}, z = \frac{25}{10}$

c. $x = \frac{17}{10}, y = \frac{-18}{10}, z = 0$

d. $x = \frac{17}{10}, y = \frac{-411}{200}, z = \frac{8087}{4000}$

6. If $Z[f(k)] = F(z)$ then $Z[f(k \pm n)] =$ _____.

a. $z^{\pm n} F(z)$

b. $z^{+n} F(z)$

c. $z^{-n} F(z)$

d. none of these

7. The False position method formula to find approximate root of the equation $f(x)=0$ in the interval $[a,b]$ is _____.

a. $c = a - \frac{f(a)}{f(b) - f(a)}(b - a)$

b. $c = a - \frac{f(a)}{f(b) + f(a)}(b - a)$

c. $c = a - \frac{f(a)}{f(b) - f(a)}(b + a)$

d. $c = a + \frac{f(a)}{f(b) - f(a)}(b - a)$

Q.5 Attempt any four questions.

[16]

- (i) Solve the equation $2x - 3 \sin x - 5 = 0$ for the root lying between 2 and 3 using Bisection method, correct up to two decimal digits.
- (ii) Solve the equation $\frac{dy}{dx} = \frac{-y}{1+x}$, $y(0.3)=2$, correct to four decimal places, use Euler's method and find $y(1)$. Take $h = 0.1$.
- (iii) Find the Z-transformation of the following (a) e^{an} (b) ne^{an} .
- (iv) Using Lagrange's interpolation formula, Compute $f(2)$ from the following table

x	0	1	3	4
y	5	6	50	105
$f(x)$				

- (v) Find the value of $\int_0^1 \frac{dx}{1+x^2}$, taking 5 sub-intervals, by Trapezoidal rule, correct to 5 significant figures. Also compare it with exact value.
- (vi) Compute the positive root of $x^3 - x - 0.1 = 0$, by Newton -Raphson Method up to six decimal places. Take $x_0 = 1$.

Q.6(a) Attempt any two questions.

[06]

- (i) Use convolution theorem to evaluate $Z^{-1}\left\{\frac{z^2}{(z-a)(z-b)}\right\}$.
- (ii) Find a real root of the equation $3x - \cos x - 1 = 0$, by Regula-Falsi method, correct to four decimal figures in the interval $[0, 1]$

- (iii) The observed the values of the function are respectively 2, 1, 0 and -1 at the four positions $-1, 0, 1$ and 3 at the independent variable, Use Newtons divided difference formula to find $f(x)$ as a polynomial in x .

(b) **Attempt any one question.**

[06]

- (i) Use the table value to find $\log_{10} 2.02$ and $\log_{10} 2.91$, by interpolation formula

x	2.0	2.2	2.4	2.6	2.8	3.0
$f(x) = \log_{10} x$	0.30103	0.34242	0.38021	0.41497	0.44716	0.47721

- (ii) Find the inverse Z-transform of $\frac{1}{(z-3)(z-2)}$,

(i) $|z| < 2$ (ii) $2 < |z| < 3$.

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